

Systematic Localization of Late-Time Cosmological Signals in Modified Gravity: CMB Survival, the Lensing Barrier, and a Growth-Sector Sign Constraint

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Abstract. We present a systematic study of where late-time cosmological signals can reside within modified-gravity (MG) parameter space, using Energy-Flow Cosmology (EFC) as a concrete case study. Starting from a late-time background modification to the Friedmann equation ($E^2 = E_{\Lambda\text{CDM}}^2 + \alpha g(a)$), we test EFC against Planck 2018 CMB likelihoods (TTTEEE, lowl, lensing), DESI 2024 BAO, SDSS DR12 BAO, and Pantheon+ supernovae, using CLASS and cobaya for background constraints, and MGCAMB for perturbation-sector constraints. The analysis proceeds through five phases: (i) the background gate collapses to $\alpha \approx 0$ under joint CMB+BAO constraints, with a near-perfect α - H_0 degeneracy ($\text{corr} = 0.975$); (ii) pure growth-modification ($\mu < 1$, $\Sigma = 1$) is blocked by CMB lensing; (iii) a narrow degeneracy valley is found at ($\mu \approx 0.94$, $\Sigma \approx 1.05$), yielding $\Delta\chi^2 = -0.45$ relative to GR with free cosmological parameters; (iv) this valley is shown to be degenerate with the phenomenological A_L anomaly ($\cos = 0.93$); (v) growth-rate data ($f\sigma_8$) independently confirm the $\mu < 1$ direction at $> 2\sigma$, breaking the A_L degeneracy. We conclude that the EFC background sector is observationally empty, the perturbation sector survives Planck within a narrow slip window ($\mu \in [0.93, 0.96]$, $\Sigma \in [1.03, 1.07]$), and a relativistic perturbation theory producing gravitational slip ($\eta \neq 1$) is required to connect the framework to cosmological data.

Keywords: modified gravity, CMB constraints, gravitational slip, $f\sigma_8$, A_L anomaly, dark energy perturbations

1. Introduction

Testing modified gravity (MG) against cosmic microwave background (CMB) data is the most demanding consistency check in cosmology. Most alternative frameworks fail because they modify early-universe physics, disrupting the acoustic peak structure, baryon loading, or the sound horizon r_s . Frameworks that confine their modifications to late times face a different challenge: the CMB is sensitive to late-time physics only through the integrated Sachs–Wolfe (ISW) effect and gravitational lensing of the primary anisotropies.

Energy-Flow Cosmology (EFC) is a framework based on the ontological primitives *grid*, *energy flow*, and *entropy* (grid+EF+S), in which gravitational dynamics emerge from entropy gradients on a discrete spatial structure [1, 2]. Previous work established consistency at galactic scales (SPARC rotation curves, Bullet Cluster) and cluster scales (entropy profiles), using a Grid-AQUAL numerical solver that passed Newton-limit and MOND-limit kill-tests.

In this note we present the first systematic CMB constraints on EFC, organized as a *localization study*: rather than fitting a single model, we progressively eliminate channels (background, pure growth, Horndeski friction) until the surviving signal region is precisely identified. This approach yields results that are largely model-independent and applicable to any late-time MG framework with similar structure.

Throughout, we use Planck 2018 likelihoods [3], DESI 2024 BAO [4], SDSS DR12 BAO, and Pantheon+ supernovae [5]. Boltzmann calculations use CLASS v3.3.4 [6] with EFC modifications (EFCLASS) for background constraints, and MGCAMB v1.5.2 [8] for perturbation-sector constraints. Parameter inference uses cobaya [7].

2. Phase 1: Background Gate Constraints

2.1 The EFC background modification

EFC modifies the Friedmann equation through an additive gate:

$$E^2(a) = E_{\Lambda\text{CDM}}^2(a) + \alpha g(a), \quad (1)$$

where $g(a) = [1 + \exp(-(a - a_t)/\delta_a)]^{-1}$ is a logistic gate activating at scale factor a_t with width δ_a , and α parameterizes the modification strength. Late-time pipeline fits to BAO, $H(z)$, and $f\sigma_8$ data had yielded $\alpha \approx -0.7$ [2].

2.2 Planck-only: the α – H_0 degeneracy

We perform a minimize scan over 11 values of $\alpha \in [0, -0.7]$ with 7 free cosmological parameters (H_0 , ω_b , ω_{cdm} , n_s , τ , $\ln(10^{10} A_s)$, A_{planck}) against the full Planck 2018 likelihood.

The result is striking: $|\Delta\chi^2| < 2$ across the entire α range. The CMB is effectively blind to α because the angular acoustic scale $\theta_* = r_s/D_A$ is preserved by compensating shifts in H_0 . Notably, the Planck-only best-fit with α free lands at $\alpha_{\text{bf}} = +0.035$ (Table 1), indicating that CMB alone does not even pull toward negative α .

MCMC with α free (Level 3, 2399 samples) confirms:

$$\alpha = -0.32 \pm 0.40 \quad (0.82\sigma), \quad \text{corr}(\alpha, H_0) = 0.975. \quad (2)$$

Zero is well within the 95% credible interval.

2.3 Joint fit: BAO breaks the degeneracy

Adding geometric probes (BAO, SNe) that directly measure $D_A(z)$ and $H(z)$ breaks the α – H_0 corridor.

Table 1: Joint-fit $\Delta\chi^2$ (EFC α -free vs. ΛCDM , 1 extra DOF).

Dataset	$\chi_{\Lambda\text{CDM}}^2$	χ_{EFC}^2	α_{bf}	$\Delta\chi^2$
Planck only	1010.35	1012.23	+0.035	+1.88
Planck + SDSS DR12	1016.91	1015.60	−0.111	−1.31
Planck + DESI 2024	1025.65	1028.63	−0.045	+2.98
Planck + DESI + Pantheon+	2432.71	2430.61	−0.045	−2.10

No dataset combination exceeds 2σ significance for the extra parameter. We find no evidence for a nonzero α in any joint fit; all results are consistent with $\alpha = 0$.

2.4 Structural conclusion

The late-time signal $\alpha \approx -0.7$ from BAO/ $H(z)$ / $f\sigma_8$ pipelines was an effective parameter absorbing physics that resides elsewhere. In the absence of any background-level signal, the perturbation sector becomes the natural candidate: modifications to the Poisson equation (μ), the lensing potential (Σ), or gravitational slip ($\eta = \Phi/\Psi \neq 1$).

3. Phase 2: Perturbation-Sector Constraints

3.1 Pure μ modification ($\Sigma = 1$)

Using MGCAMB with constant μ across all redshift bins and $\Sigma = 1$ fixed, we scan $\mu \in [0.80, 1.05]$ against Planck 2018.

Table 2: Pure μ scan ($\Sigma = 1$ fixed). $\Delta\chi^2$ relative to GR.

μ	$\Delta\sigma_8$	$\Delta\chi_{\text{TT}}^2$	$\Delta\chi_{\text{lens}}^2$	$\Delta\chi_{\text{full}}^2$	Verdict
0.90	−10.3%	+18.5	+14.7	+33.2	Excluded
0.93	−7.3%	+11.5	+7.5	+19.0	Excluded
0.95	−5.3%	+7.6	+3.9	+11.5	Excluded
0.97	−3.2%	+4.1	+1.4	+5.6	Marginal
0.99	−1.1%	+1.2	+0.2	+1.4	Viable

The σ_8 suppression needed for S_8 -tension relief ($\sim 8\%$) requires $\mu \approx 0.93$, but this incurs $\Delta\chi^2 = +19$. CMB lensing ($C_\ell^{\phi\phi}$) is *not* blind to μ , contributing roughly half the total penalty. Pure growth-modification with $\Sigma = 1$ is excluded.

3.2 The μ – Σ degeneracy valley

Releasing Σ as a free parameter reveals a narrow valley where $\Sigma > 1$ compensates the lensing penalty from $\mu < 1$. We scan a 25-point grid ($\mu \in [0.92, 1.00]$, $\Sigma \in [0.90, 1.10]$) at fixed cosmological parameters, followed by a finer 36-point grid with per-component likelihood decomposition.

Table 3: Profiled valley $\Sigma^*(\mu)$ from the 2D scan (fixed cosmological parameters).

μ	Σ^*	$\Delta\chi_{\text{full}}^2$	$\Delta\sigma_8$	S_8 relief
0.92	1.060	+4.56	−8.4%	100%
0.93	1.050	+3.45	−7.4%	93%
0.94	1.050	+2.49	−6.4%	81%
0.95	1.040	+1.61	−5.3%	67%

A parabolic fit to the lensing likelihood at $\mu = 0.94$ gives $\Sigma_{\text{min}} = 1.047 \pm 0.013$. The valley is narrow ($\sigma_\Sigma = 0.013$) but well-defined.

3.3 Minimize with free cosmological parameters

Running cobaya minimize (BOBYQA, best-of-2) at three anchor points with 7 free parameters (H_0 , ω_b , ω_{cdm} , n_s , τ , $\ln(10^{10}A_s)$, A_{planck}) at fixed μ and Σ yields:

Table 4: Minimize results with free cosmological parameters. $\Delta\chi^2$ relative to GR ($\mu = 1$, $\Sigma = 1$, $\chi^2 = 1014.78$).

μ	Σ	χ_{min}^2	$\Delta\chi^2$	$\Delta\chi_{\text{highl}}^2$	$\Delta\chi_{\text{lowlTT}}^2$
0.94	1.05	1014.33	−0.45	−4.68	+2.97
0.95	1.04	1015.90	+1.13	−0.40	+0.93
0.92	1.06	1019.90	+5.12	−5.48	+5.91

At the sweet spot ($\mu = 0.94$, $\Sigma = 1.05$), the high- ℓ likelihood *improves* by $\Delta\chi^2 = -4.68$, compensating the ISW penalty (+2.97) and lensing cost (+0.41). GR is not preferred; the net improvement is $\Delta\chi^2 = -0.45$ for 2 extra parameters. However, this is not statistically significant (requires $|\Delta\chi^2| > 5.99$ for 2σ with 2 DOF).

3.4 Component anatomy

The ISW decomposition reveals the mechanism:

- **High- ℓ (plik):** Improves because $\Sigma > 1$ enhances lensing-smoothing of acoustic peaks—precisely the effect parameterized by Planck’s phenomenological $A_L > 1$.
- **Low- ℓ TT (ISW):** Penalizes because $\mu < 1$ modifies the late-time Poisson equation, altering the ISW signal. This is a monotonically increasing cost.
- **Lensing ($C_\ell^{\phi\phi}$):** Near-minimum at $\Sigma \approx 1.047$, with a sharp parabolic profile.
- **Low- ℓ EE:** Negligible sensitivity to (μ, Σ) .

4. Phase 3: The A_L Degeneracy

The high- ℓ improvement at $\Sigma > 1$ raises a critical question: is the (μ, Σ) valley physically distinct from the known $A_L > 1$ anomaly in Planck data?

We perform four independent diagnostic tests.

4.1 Test 1: χ^2 comparison

Table 5: A_L degeneracy diagnostic.

Model	$\chi^2_{\min}$	$\Delta\chi^2$	A_L^{bf}	DOF
GR reference	1014.78	0	1.00	0
GR + A_L free	1010.36	-4.42	1.042	1
$(\mu, \Sigma) + A_L = 1$	1014.33	-0.45	1.00	2
$(\mu, \Sigma) + A_L$ free	1016.37	+1.59	0.976	3

When MG is active, the optimizer drives A_L *below* 1 (0.976), and total χ^2 worsens by 6.0 units relative to A_L alone. The two effects compete for the same likelihood improvement.

4.2 Test 2: Fisher matrix

The Fisher matrix in (μ, Σ, A_L) space has a flattest eigenvector satisfying $\cos(\hat{e}_{\text{flat}}, \hat{A}_L) = 0.81$, confirming near-perfect degeneracy along the A_L direction.

4.3 Test 3: Gradient decomposition

At $(\mu = 0.94, \Sigma = 1.05)$, the per-component gradients satisfy $\cos(\nabla_{\Sigma}\chi^2, \nabla_{A_L}\chi^2) = 0.93$: Σ and A_L produce identical χ^2 patterns in high- ℓ and lensing (ratio = 1.63 in both components). The only distinction: Σ incurs an ISW cost (+64.1) while A_L has zero ISW sensitivity.

4.4 Conclusion

Three independent diagnostics converge: the (μ, Σ) valley is the A_L anomaly expressed through physical metric potentials. $\Sigma > 1$ is the *physical implementation* of $A_L > 1$: both modify lensing-smoothing of the primary power spectrum identically. CMB data alone cannot distinguish them.

5. Phase 4: Growth-Sector Sign Constraint

The A_L degeneracy renders CMB-only constraints insufficient. However, growth-rate data ($f\sigma_8$) measure structure formation directly and are insensitive to A_L .

5.1 Bridge test B0

We test $\mu < 1$ and $\mu > 1$ against a compilation of 7 $f\sigma_8$ measurements ($z = 0.38\text{--}0.85$) using modified screening filters with $\alpha_{\text{eff}} = 0.06$ (6% effective modification):

Table 6: B0 bridge test: $f\sigma_8$ response to modified gravity.

Filter	Direction	$\Delta\sigma_8$	χ^2	$\Delta\chi^2$	Verdict
Λ CDM	—	—	66.62	—	Reference
F3 tanh	Weaken	−0.54%	61.40	−5.22	$> 2\sigma$ impr.
F2 gauss	Weaken	−0.25%	63.62	−3.00	Improves
F1 raised	Weaken	−0.18%	64.45	−2.17	Improves
F0 original	Weaken	−0.00%	66.61	−0.01	Neutral
F1 raised	Strengthen	+16.8%	561.4	+494.7	Killed

The sign constraint is unambiguous: $\mu < 1$ (weakened gravity) systematically improves $f\sigma_8$, while $\mu > 1$ is excluded with $\Delta\chi^2 > 400$. The best-performing filter (F3, tanh-band reaching $k \sim 0.05\text{--}0.1\ h/\text{Mpc}$) yields $\Delta\chi^2 = -5.22$ for 1 DOF—a $> 2\sigma$ improvement.

Critically, this result is *independent of A_L* : $f\sigma_8$ measures growth directly and has no sensitivity to CMB lensing-smoothing. The $\mu < 1$ preference from growth data thus breaks the A_L degeneracy identified in Phase 3.

5.2 Relevant scales

The original EFC screening scale ($k_\Lambda = 0.0014\ h/\text{Mpc}$) produces zero effect on $f\sigma_8$ (F0: $\Delta\chi^2 = -0.01$). The modification must reach $k \approx 0.05\text{--}0.1\ h/\text{Mpc}$ to affect growth at the relevant redshifts.

6. Phase 5: Theory–Observation Gap

6.1 GRAV pipeline extraction

The existing EFC numerical implementation (Grid-AQUAL) solves a single scalar Poisson equation with a MOND-like interpolating function:

$$\nabla \cdot [\mu_{\text{AQUAL}}(|\nabla\Phi|/a_0) \nabla\Phi] = 4\pi G\rho. \quad (3)$$

This solver has passed Newton-limit (KT1) and MOND-limit (KT2) kill-tests at galactic scales.

However, mapping to the cosmological (μ, Σ) parametrization reveals three structural incompatibilities:

1. **Wrong sign:** Grid-AQUAL gives $\mu_{\text{GRAV}} > 1$ (MOND strengthens gravity), while the CMB valley and $f\sigma_8$ data require $\mu < 1$.
2. **No gravitational slip:** A single potential implies $\Phi = \Psi$ ($\eta = 1$) everywhere, while the CMB valley requires $\mu \approx 0.94$ and $\Sigma \approx 1.05$ ($\eta \approx 1.12$).
3. **Fourier screening:** At $k_\Lambda = 0.0014 \, h/\text{Mpc}$, the modification is $< 0.1\%$ at CMB-relevant scales ($k > 0.01 \, h/\text{Mpc}$), far below the required $\sim 6\%$.

6.2 Implications

The galactic-scale solver and the cosmological perturbation sector occupy different regions of parameter space with different signs, different potential structures, and different scale dependencies. Closing this gap requires a *relativistic* EFC perturbation theory that produces two independent metric potentials with $\mu < 1$ and $\Sigma > 1$ at late times—analogueous to how TeVeS [9] provided a relativistic completion of MOND.

7. Discussion

7.1 Summary of constraints

Table 7: Complete constraint summary.

Test	Result	Significance
Background α (Planck)	Blind	$\text{corr}(\alpha, H_0) = 0.975$
Joint α (Planck+BAO+SN)	$\alpha \rightarrow -0.045$	Background gate empty
Pure μ ($\Sigma = 1$)	Excluded	Lensing blocks S_8 relief
μ - Σ valley	$\Delta\chi^2 = -0.45$	Viable but degenerate with A_L
A_L diagnostic	$\cos = 0.93$	Same CMB modes as A_L
$f\sigma_8$ (B0)	$\Delta\chi^2 = -5.22$	$\mu < 1$ at $> 2\sigma$, breaks A_L
GRAV $\rightarrow (\mu, \Sigma)$	Does not close	Wrong sign, no slip, screening

7.2 The surviving signal specification

The combined analysis yields a precise specification for any late-time MG perturbation theory:

- **Sign:** $\mu < 1$ (weakened matter-potential coupling). Locked by $f\sigma_8$ independently of CMB.
- **Magnitude:** $\sim 6\%$ effective weakening ($\mu \approx 0.94$).
- **Lensing:** $\Sigma \approx 1.05$ to compensate CMB lensing penalty. Gravitational slip ($\eta \neq 1$) is essential.
- **Scales:** Effect must reach $k \approx 0.05\text{--}0.1\ h/\text{Mpc}$.
- **Epoch:** Late-time ($z < 1$), consistent with dark-energy-dominated era.

7.3 Connection to the A_L anomaly

The finding that $\Sigma > 1$ reproduces the $A_L > 1$ effect through physical metric modification is noteworthy independently of EFC. Planck’s A_L anomaly ($A_L = 1.042$ in our minimize) lacks a physical mechanism in ΛCDM . Gravitational slip with $\Sigma \approx 1.05$ provides one: enhanced $\Phi + \Psi$ at late times increases lensing-smoothing of the acoustic peaks without phenomenological rescaling. However, as shown in Phase 3, CMB data alone cannot distinguish $\Sigma > 1$ from $A_L > 1$.

7.4 Regime-split architecture

The structural gap between galactic-scale EFC (MOND-like, $\mu > 1$, $\eta = 1$) and cosmological EFC ($\mu < 1$, $\eta \neq 1$) points toward a regime-split architecture: the same underlying ontology (grid+EF+S) producing different effective gravitational laws at different scales and in different environments. This is consistent with the EFC framework’s emphasis on entropy-gradient-driven dynamics, where the gradient structure differs fundamentally between collapsed (galactic) and linear (cosmological) regimes.

7.5 Limitations

1. All perturbation-sector results use constant (μ, Σ) . Time-dependent parametrizations may open or close the valley.
2. The $f\sigma_8$ compilation uses 7 data points with known systematic uncertainties (galaxy bias, velocity reconstruction). The B0 bridge test is a fixed-parameter consistency check using

simplified screening filters, not a full joint-likelihood analysis with marginalization over RSD systematics. The $> 2\sigma$ improvement should be interpreted as directional evidence, not a detection.

3. No joint Planck + weak-lensing + BAO fit with free (μ, Σ) has been performed. This is the critical next step.
4. The EFC perturbation theory that would predict specific $\mu(a, k)$ and $\Sigma(a, k)$ does not yet exist.

8. Reproducibility

All computations were performed on two workstations: a desktop PC (Intel i9-14900K, 2× NVIDIA RTX 5080, 192 GB RAM, WSL2/Ubuntu 24) and a Mac Studio (Apple M3 Ultra, 512 GB RAM, macOS). Background constraints: CLASS v3.3.4 with EFC modifications (EFCCLASS), cobaya 3.6.1. Perturbation-sector constraints: MGCAMB v1.5.2 (MG_flag=6, spline reconstruction), cobaya 3.6.1. Planck 2018 likelihoods: plik_lite TTTEEE, lowl_TT, lowl_EE, lensing. BAO: DESI 2024, SDSS DR12. Supernovae: Pantheon+. Configuration files and analysis scripts are available upon request.

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